Solutions to the final exam paper on December 16, 2022

1. (a)

$$\begin{aligned}\mathbb{E}(S_n \mid X_1) &= \mathbb{E}(X_1 + X_2 + \cdots + X_n \mid X_1) = \\ &= \mathbb{E}(X_1 \mid X_1) + \mathbb{E}(X_2 \mid X_1) + \cdots + \mathbb{E}(X_n \mid X_1) \\ &= X_1 + \mathbb{E}(X_2) + \cdots + \mathbb{E}(X_n) = X_1 + (n-1)\lambda. \end{aligned} \quad (3 \text{ pts})$$

Grading criteria: 2 points can be given if one specify $\mathbb{E}(X_1|X_1) = X_1$ instead of λ.

(b) By symmetry, the n conditional expectations $\mathbb{E}(X_i \mid S_n = s)$ are all equal, and sum up to s. So $\mathbb{E}(X_1 \mid S_n = s) = s/n$, and $\mathbb{E}(X_1 \mid S_n) = S_n/n$.

(c) Given $S_n = s \in \mathbb{N}$,
(c1) $(X_1, \ldots, X_n)$ take values in

$$D = \{(x_1, \ldots, x_n) \mid x_j \in \mathbb{N}, \quad x_1 + \cdots + x_n = s\}; \quad (2 \text{ pts})$$

Grading criteria: 1 points can be given if one specify both $x_j \geq 0$ and $\sum_j x_j = s$.
(c2) Note that S_n has the Poisson distribution with parameter $n\lambda$. For $(x_1, \ldots, x_n) \in D$,

$$\begin{aligned}&\mathbb{P}\left[(X_1, \ldots, X_n) = (x_1, \ldots, x_n) \mid S_n = s\right] \\ &= \frac{\mathbb{P}\left[(X_1, \ldots, X_n) = (x_1, \ldots, x_n), S_n = s\right]}{\mathbb{P}(S_n = s)} \\ &= \frac{\mathbb{P}\left[(X_1, \ldots, X_n) = (x_1, \ldots, x_n)\right]}{\mathbb{P}(S_n = s)} \\ &= \frac{e^{-\lambda n}\lambda^s/(x_1! \cdots x_n!)}{e^{-n\lambda}(n\lambda)^s/s!} \quad (1 \text{ pts}) \\ &= \frac{s!}{x_1! \cdots x_n!\, n^s}. \quad (1 \text{ pts})\end{aligned}$$

2. (a) We have $X_n = \min(Z_0, Z_1, \ldots, Z_n), n \geq 1$
The transition matrix shows as follows:

$$\mathbf{P} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0.1 & 0.9 & 0 & 0 \\ 0.1 & 0.3 & 0.6 & 0 \\ 0.1 & 0.3 & 0.2 & 0.4 \end{bmatrix}$$

(b) $\mathbb{P}(X_3 = 2, X_2 = 1, X_1 = 2 | X_0 = 3) = P_{32}P_{21}P_{12} = 0$

(c) $\mathbb{P}(X_3 = 0, X_2 = 1, X_1 = 2 | X_0 = 3) = P_{32}P_{21}P_{10} = 0.2 * 0.3 * 0.1 = 0.006$

(d) Since $X_m \leq X_{n-1} (\forall m \geq 1)$,
$i \to j$ is impossible if $i < j$.
Therefore, all classes singleton and we have 4 classes: $0, 1, 2, 3$

(e) For $i = 1, 2, 3, P_{i,i-1} > 0$ and once at $i - 1$, the MC cannot return to i.
Therefore $f_i < 1$ and i is transient.
For $i = 0$ is absorbing, thus recurrent.

(f) For all i, $p_{ii} > 0$, so "i" is aperiodic.

(g) For $\pi = (\pi_0, \pi_1, \pi_2, \pi_3)$ Solving $\pi P = \pi$, i.e.

$$\begin{cases} \pi_0 = \pi_0 + 0.1\pi_1 + 0.1\pi_2 + 0.1\pi_3 \\ \pi_1 = 0.9\pi_1 + 0.3\pi_2 + 0.3\pi_3 \\ \pi_2 = 0.6\pi_2 + 0.2\pi_3 \\ \pi_3 = 0.4\pi_3 \end{cases}$$

From the equations: $\pi_3 = 0$, then $\pi_2 = 0, \pi_1 = 0, \pi_0 = 1$
There is one unique stationary distribution$\pi = (1, 0, 0, 0)$

(h) Since $j = 1, 2, 3$ are transient,
$P_{ij}^n \to 0$ as $n \to \infty$ for all $0 \le i \le 3$ and $1 \le j \le 3$
Moreover,

$$1 = P_{i0}^n + P_{i1}^n + P_{i2}^n + P_{i3}^n$$

implies $P_{i0}^n \to 1 (\forall 0 \le i \le 3)$
Therefore, as $n \to \infty$

$$P^n \to \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

which are the limiting probability of the MC.

Grading criteria:

(a) 4pts for correct answers, 2pts for partially correct answers (e.g., some rows of the transition matrix are correct), and no points for completely incorrect answers.

(b) Full marks for correct answers, no points for incorrect answers.

(c) Full marks for correct answers, no points for incorrect answers.

(d) Full marks for correct answers, no points for incorrect answers.

(e) 1pt for $i = 0$ is transient, another 1pt for $i = 1, 2, 3$ are recurrent.

(f) Full marks for correct answers, no points for incorrect answers.

(g) 1pts for the balance equations and 2pts for the correct answer.

(h) 3pts for correct answers, 2pts for partially correct answers (e.g., give some (not all) of the limiting probabilities), and no points for completely incorrect answers.

3. (a)

$$\begin{aligned} &\mathbb{P}(X_1 < X_2 < X_3) \\ =&\mathbb{E}\left[1_{\{X_1<X_2\}} 1_{\{X_2<X_3\}}\right] && \text{(2 pts)} \\ =&\mathbb{E}\left[1 - e^{-\lambda_1 X_2} e^{-\lambda_3 X_2}\right] \qquad \text{(by conditioning on } X_2\text{)} && \text{(3 pts)} \\ =&\mathbb{E}\left[e^{-\lambda_3 X_2}\right] - \mathbb{E}\left[e^{-(\lambda_1+\lambda_3)X_2}\right] \\ =&\frac{\lambda_2}{\lambda_2+\lambda_3} - \frac{\lambda_2}{\lambda_2 + +\lambda_1 + \lambda_3} = \frac{\lambda_1\lambda_2}{(\lambda_2+\lambda_3)(\lambda_1+\lambda_2+\lambda_3)}. \end{aligned}$$

Grading criteria: 5 points can be given for other methods like conditioning probability-based method and integral-based method. However,

- 5 points is given if you take conditional probability like $\mathbb{P}(X_1 < X_2 < X_3) = \mathbb{P}(X_2 < X_3 | X_1 < \min\{X_2, X_3\})\, \mathbb{P}(X_1 < \min\{X_2, X_3\})$ and clearly quote the "**strong memoryless property**".
- 3 points is given if you take conditional probability like $\mathbb{P}(X_1 < X_2 < X_3) = \mathbb{P}(X_2 < X_3 | X_1 < \min\{X_2, X_3\})\, \mathbb{P}(X_1 < \min\{X_2, X_3\})$ without quoting the "**strong memoryless property**".
- 0 points if you take condition probability like $\mathbb{P}(X_1 < X_2 < X_3) = \mathbb{P}(X_1 < \min\{X_2, X_3\} | X_2 < X_3)\, \mathbb{P}(X_2 < X_3)$ (probably unworkable)
- 0 points if you simply use $\mathbb{P}(X_1 < X_2 < X_3) = \mathbb{P}(X_1 < \min\{X_2, X_3\}) \cdot \mathbb{P}(X_2 < X_3)$ without proving their independence (also probably unworkable).

(b) When all the λ_j's are equal, we find

$$\mathbb{P}(X_1 < X_2 < X_3) = \frac{1}{2 \cdot 3} = \frac{1}{3!}. \qquad \text{(2 pts)}$$

This can be also found as follows. Here the 3 variables are i.i.d. exponentially distributed. By symmetry, all the 3! ranking of the variables,

$$X_i < X_j < X_k,$$

where (i, j, k) is a permutation of $\{1, 2, 3\}$, are equally possible. Therefore, $\mathbb{P}(X_1 < X_2 < X_3) = 1/3!$

Grading criteria: The remaining 3 points can be given if one specify some key ideas like *permutation* and *order statistics*. Students may also be given points if they succeeded using some method that is different from one in 3(a).

4. Let $\{N_t\}_{t \geq 0}$ be a Poisson process with rate 3.

(a) $N_4 \sim \mathcal{P}(12)$ so that $P(N_4 = 3) = e^{-12}\frac{12^3}{3!} \approx 0.00177$.

(b)

$$\begin{aligned} &P(N_4 = 3, N_6 = 6, N_{10} = 12) = P(N_4 = 3, N_6 - N_4 = 3, N_{10} - N_6 = 6) \\ =&P(N_4 = 3)P(N_6 - N_4 = 3)P(N_{10} - N_6 = 6) \\ =&e^{-12}\frac{12^3}{3!} \times e^{-6}\frac{6^3}{3!} \times e^{-12}\frac{12^6}{6!} \\ =&e^{-30}\frac{12^8}{10} \approx 4 \cdot 10^{-6}. \end{aligned}$$

(c) We have $N_6 = N_4 + (N_6 - N_4)$ where the two summands are independent.

$$\begin{aligned} P(N_6 = 6|N_4 = 3) = \frac{P(N_4 = 3, N_6 - N_4 = 3)}{P(N_4 = 3)} &= \frac{P(N_4 = 3)P(N_6 - N_4 = 3)}{P(N_4 = 3)} \\ &= P(N_6 - N_4 = 3) = e^{-6}\frac{6^3}{3!} = e^{-6}6^2 \approx 0.0892 \end{aligned}$$

(d) As in (c),

$$\begin{aligned} P(N_6 = 6|N_{10} = 12) &= \frac{P(N_{10} = 12|N_{N_6} = 6)\ P(N_6 = 6)}{P(N_{10} = 12)} \\ &= \frac{P(N_{10} - N_{N_6} = 6)\ P(N_6 = 6)}{P(N_{10} = 12)} \\ &= \binom{12}{6}\left(\frac{3}{5}\right)^6\left(\frac{2}{5}\right)^6 \approx 0.1766. \end{aligned}$$

(e) We have $N_{10} = N_6 + (N_{10} - N_6)$ where the two summands are independent. Thus

$$E[N_{10}|N_6] = E[N_6 + (N_{10} - N_6)|N_6] = N_6 + E[(N_{10} - N_6)|N_6] = N_6 + E[N_{10} - N_6] = N_6 + 12.$$

Similarly, observing that $(N_{10} - N_6)$ is independent of (N_4, N_6),

$$\begin{aligned} &E[N_{10}|N_4, N_6] = E[N_6 + (N_{10} - N_6)|N_4, N_6] = N_6 + E[(N_{10} - N_6)|N_4, N_6] \\ &= N_6 + E[N_{10} - N_6] = N_6 + 12. \end{aligned}$$

(f) Assume $N_{10} = n$. Then N_6 takes values in $\{0, \dots, n\}$. Repeating the calculation of (d), we find for $0 \le x \le n$,

$$P(N_6 = x | N_{10} = n) = \binom{n}{x} \left(\frac{3}{5}\right)^x \left(\frac{2}{5}\right)^{n-x}.$$

which is the requested binomial distribution.

It follows that $E(N_6|N_{10} = n) = \frac{3}{5}n$ and thus $\boxed{E(N_6|N_{10}) = \frac{3}{5}N_{10}}$.

- Rules

  (a) $\mathcal{P}(12)$ 1pt, answer 1 pt

  (b)-(d) The correct decomposition 1pt, here you need to use the property of possion process to do decompose, only the bayes rule will be graded as 0. Correct answer 1 pt

  (e) Each expectation 1 pt

  (f) The proof part 3 pt and the expectation 2 pt

5. (a) Arrival time of the n-th claim in portfolio $X = S_{x,n}$

$$\begin{aligned}
\mathbb{P}(S_{A,3} < S_{B,3}) &= \sum_{k=3}^{3+3-1} \binom{3+3-1}{k} (\frac{3}{3+5})^k (\frac{5}{3+5})^{3+3-1-k} && \text{(3 pts)} \\
&= \sum_{k=3}^{5} \binom{5}{k} (\frac{3}{8})^k (\frac{5}{8})^{5-k} && \text{(1 pts)} \\
&= 0.2752 && \text{(1 pts)}
\end{aligned}$$

- At most 3 pts is given for wrong answers like $\sum_{k=3}^{5} \binom{5}{k} (\frac{3}{5})^k (\frac{5}{8})^{5-k}$, $\sum_{k=3}^{5} \binom{5}{k} (\frac{5}{8})^k (\frac{3}{8})^{5-k}$ or $\binom{5}{3} (\frac{3}{8})^3 (\frac{5}{8})^{5-3}$.
- At mot 2 pts is given for wrong answers containing e.

(b) The supposed claim process $\{N(t)\}_{t \ge 0}$ is a Poisson process with rate $\lambda = 8$ per year. (1 pts)
Given that $N(1) = 8$, the arrival time are distributed as the order statistics of $U_1, \dots, U_8 \overset{iid}{\sim} U(0, 1)$. (2 pts. No score will be given for exponential distribution.)
In particular, $U_{(1)} = \min U_1, \dots, U_n$

$$\begin{aligned}
\mathbb{P}(U_{(1)} > t) &= \mathbb{P}(U_1 > t, \dots, U_8 > t) \\
&= \prod_{i=1}^{8} \mathbb{P}(U_i > 8) \\
&= (1-t)^8, t > 0 && \text{(1 pts)}
\end{aligned}$$

Therefore, the conditional expected arrival time of S_1 is

$$\mathbb{E}(S_1|N(1) = 8) = \mathbb{E}(U_{(1)}) = \int_0^1 (1-t)^8 dt = \frac{1}{9} \quad \text{(1 pts)}$$

- 0 pts is given if you simply write $\mathbb{E}(S_1|N(1) = 8) = \frac{1}{8}$ without further analysis.

- 3 pts is given if anything else is right but wrongly write $\mathbb{P}(U_{(1)} > t) = (1-t)^7$.

6. (a) To prove that X is irreducible, For we need to prove that for any state (n_1, n_2), it can reach any other state (m_1, m_2) with a positive probability.
We prove this statement by proving that (n_1, n_2) can reach state $(0,0)$ and that state $(0,0)$ can reach state (m_1, m_2). We first prove that state (n_1, n_2) can reach state $(0,0)$. We illustrate this proof by considering state $(n_1, n_2) = (2,3)$. The following diagram proves that state $(2,3)$ can reach state $(0,0)$ in positive probability:

$$(2,3) \overset{\mu_1}{\to} (1,4) \overset{\mu_1}{\to} (0,5) \overset{.8\mu_2}{\to} (0,4) \overset{.8\mu_2}{\to} (0,3) \overset{.8\mu_2}{\to} (0,2) \overset{.8\mu_2}{\to} (0,1) \overset{.8\mu_2}{\to} (0,0)$$

The following diagram illustrates the proof that state $(0,0)$ can reach state (m_1, m_2) by choosing $(m_1, m_2) = (2,1)$:

$$(0,0) \overset{\alpha}{\to} (1,0) \overset{\alpha}{\to} (2,0) \overset{\alpha}{\to} (3,0) \overset{\mu_1}{\to} (2,1).$$

- 5 pts is given if you draw the diagram of $(m_1, m_2), \forall m_1, m_2 \geq 0$ or take the similar argument above (a simpler way is also appreciated).
- At most 3 pts is given if you analyze based on $X_1(t)$ or $X_2(t)$ instead of $(X_1(t), X_2(t))$, even if you analyze the positive recurrence of $X_1(t)$ and $X_2(t)$.
- At most 1 pts is given if you only use the following argument: "since all states communicate with each other, X are irreducible", where you just copy the definition of irreducibility but do not point out the transition between states.

(b) Let $\pi = \{\pi(n_1, n_2) : n_1, n_2 \in \mathbb{Z}_+\}$ be the stationary distribution.

(i) For each state (n_1, n_2), π must satisfies the following balance equation: the flow rate out of the state is equal to the flow rate into the state. We consider for sets of states.
(1) $(n_1, n_2) = (0,0)$: The balance equation is

$$\alpha\pi(0,0) = .8\mu_2\pi(0,1). \tag{1}$$

(2) $(n_1, n_2) = (n_1, 0)$ with $n_1 > 0$: The balance equation is

$$(\alpha + \mu_1)\pi(n_1, 0) = \alpha\pi(n_1 - 1, 0) + .8\mu_2\pi(n_1, 1) + .2\mu_2\pi(n_1 - 1, 1). \tag{2}$$

(3) $(n_1, n_2) = (0, n_2)$ with $n_2 > 0$: The balance equation is

$$(\alpha + \mu_2)\pi(0, n_2) = \mu_1\pi(1, n_2 - 1) + .8\mu_2\pi(0, n_2 + 1). \tag{3}$$

(4) $(n_1, n_2) = (n_1, n_2)$ with $n_1 > 0$ and $n_2 > 0$: The balance equation is

$$\begin{aligned}(\alpha + \mu_1 + \mu_2)\pi(n_1, n_2) \quad = \quad & \alpha\pi(n_1 - 1, n_2) + \mu_1\pi(n_1 + 1, n_2 - 1) \\ & + .8\mu_2\pi(n_1, n_2 + 1) + .2\mu_2\pi(n_1 - 1, n_2 + 1).\end{aligned} \tag{4}$$

- 4 pts in total with 1 pts for each correct balance equation.
- At most 1 pts is given in total if you miss α in the left-hand side.
- No score is given if you write left-hand side like $\alpha + 0.2\mu_1\pi(n_1, 0) =$, $\alpha + 0.2\mu_1 + 0.8\mu_2\pi(n_1, n_2)$.
- No score is given if you write like $\alpha_1 + \mu_1 = 0.8\mu_2$.
- No score is given if you write like $\alpha\pi(n_1, n_2) = \mu_1\pi(0, n_2) + \mu_2\pi(n_1, 0) + \pi(0,0)$.

(ii) We conjecture that

$$\pi(n_1, n_2) = (1 - \rho_1)\rho_1^{n_1}(1 - \rho_2)\rho_2^{n_2} \quad \text{for } n_1, n_2 \in \mathbb{Z}_+, \tag{5}$$

where $\rho_i = \lambda_i/\mu_i$, $i = 1, 2$, and (λ_1, λ_2) satisfies

$$\begin{aligned}\lambda_1 &= \alpha + .2\lambda_2, \\ \lambda_2 &= \lambda_1.\end{aligned}$$

Solving the linear system of equations we have $\lambda_1 = \lambda_2 = 1.25\alpha$.

- 2 pts for $\lambda_1 = \lambda_2 = 1.25\alpha$.
- 2 pts for correct $\pi(n_1, n_2)$.

(iii) We now verify that π, defined in (5), satisfies (1)-(4). We first verify (4). Equation (4) is equivalent to

$$\begin{aligned}(\alpha+\mu_1+\mu_2)(1-\rho_1)\rho_1^{n_1}(1-\rho_2)\rho_2^{n_2} &= \alpha(1-\rho_1)\rho_1^{n_1-1}(1-\rho_2)\rho_2^{n_2}\\ &+\mu_1(1-\rho_1)\rho_1^{n_1+1}(1-\rho_2)\rho_2^{n_2-1}+.8\mu_2(1-\rho_1)\rho_1^{n_1}(1-\rho_2)\rho_2^{n_2+1}\\ &+.2\mu_2(1-\rho_1)\rho_1^{n_1-1}(1-\rho_2)\rho_2^{n_2+1},\end{aligned}$$

which is further equivalent to

$$(\alpha+\mu_1+\mu_2)\rho_1\rho_2 = \alpha\rho_2+\mu_1\rho_1^2+.8\mu_2\rho_1\rho_2^2+.2\mu_2\rho_2^2.$$

Since $\lambda_1 = \lambda_2$, $\rho_i = \lambda_1/\mu_i$ for $i = 1, 2$. Therefore, the preceding equation is equivalent to

$$(\alpha+\mu_1+\mu_2)(\lambda_1/\mu_1)(\lambda_1/\mu_2) = \alpha(\lambda_1/\mu_2)+\mu_1(\lambda_1/\mu_1)^2+.8\mu_2(\lambda_1/\mu_1)(\lambda_1/\mu_2)^2+.2\mu_2(\lambda_1/\mu_2)^2,$$

or

$$(\alpha+\mu_1+\mu_2)\lambda_1 = \alpha\mu_1+\lambda_1\mu_2+.8\lambda_1^2+.2\lambda_1\mu_1,$$

or

$$(\alpha+\mu_1)\lambda_1 = \alpha\mu_1+.8\lambda_1^2+.2\lambda_1\mu_1.$$

Since $\lambda_1 = \alpha+.2\lambda_1$, we have $\lambda_1\mu_1 = \alpha\mu_1+.2\lambda_1\mu_1$. Thus the preceding equation is reduced to

$$\alpha\lambda_1 = .8\lambda_1^2,$$

which holds because $\alpha = .8\lambda_1$.
Equations (1)-(3) can be verified similarly.

- 2 pts for part (iii).
- At most 1 pts is given if the stationary distribution correct but some balance equations are wrong.
- No score is given if the stationary distribution is wrong.
- No score is given if you shortly claim that the stationary distribution obeys the balance equation without plugging it into balance equations.

7. (a) $B(2.1)|_{B(1.6)=0.5} \sim N(0.5, 2.1-1.6) \equiv N(0.5, 0.5)$ 2 points
$P(B(2.1) > 1.7|B(1.6) = 0.5) = 1-\Phi(\frac{1.7-0.5}{\sqrt{0.5}}) = 1-\Phi(1.697) = 0.0448$ 3 points

(b) $B(1.6)|_{B(2.1)=1.7} \sim N(\frac{1.6}{2.1}(1.7), \frac{1.6}{2.1}(2.1-1.6)) \equiv N(1.2952, 0.3810)$ 2 points
$P(B(1.6) \le 0.5|B(2.1) = 1.7) = \Phi(\frac{0.5-1.2952}{\sqrt{0.3810}}) = \Phi(-1.288) = 0.0989$ 3 points

(c)
$$\mathrm{Cov}\{B(t), B(s)-\frac{s}{t}B(t)\} = \mathrm{Cov}\{B(t), B(s)\}-\frac{s}{t}\mathrm{Var}\{B(t)\} = s-\frac{s}{t}t = 0 \text{ } 4points$$

Since $\big(B(t), B(s)-\frac{s}{t}B(t)\big)$ follows a bivariate normal distribution (it is a linear transformation of $(B(t), B(s))$) and they are uncorrelated, they must also be independent. 1 points

(d) $M(1)$ has the distribution of $|B(1)| = |N(0,1)|$. 1 points So

$$\mathbb{E}[M(1)] = \int_{-\infty}^{\infty}|x|\sqrt{\frac{1}{2\pi}}\exp(-\frac{x^2}{2})dx = 2\int_0^{\infty}\sqrt{\frac{1}{2\pi}}x\exp(-\frac{x^2}{2})dx = 2\sqrt{\frac{1}{2\pi}} = \sqrt{\frac{2}{\pi}} = 0.7979$$

4 points